Springs: Hooke’s Law and Simple Harmonic Motion

Answer key

Conventions used throughout

• **Hooke’s law (from the lesson):** $F_e = kx$ for the size of the spring force, where x is the stretch or compression measured from the unstretched position. The spring force on the object is a restoring force, $\vec{F}_e = -k\vec{x}$: it points back toward the unstretched position.

• **Energy:** $E_e = \frac{1}{2}kx^2$ (elastic potential), $E_k = \frac{1}{2}mv^2$ (kinetic). A is the amplitude, the largest displacement from equilibrium.

• **Signs:** forces are written as sizes, with + or − in the equation for their direction. The + direction is named in each question.

• **Solve in letters first:** each unknown is isolated as a formula before any numbers go in.

• **Final answers:** energies, spring constants and speeds are scalars. Forces and accelerations are given as a size with the direction in square brackets.

• g = 9.81 m/s^2. Lengths in cm are changed to m before they are used. Every step keeps full precision; only final answers are rounded.

ELASTIC POTENTIAL ENERGY

Deriving the equation

To derive an equation for elastic potential energy, consider the work done on an ideal spring in stretching or compressing it. The area under the line on a force-displacement graph indicates the work done.

W = area of triangle = ½bh

a) sub in from the graph in terms of F and x:

b) Now use Hooke’s law (without the negative) sub in for force in terms of k and x. Then simplify and check with your teacher

The work done to stretch the spring is the area of the triangle under the force–displacement line.

a) sub in from the graph in terms of F and x:

On the graph the base of the triangle is the stretch x and the height is the spring force F_e at that stretch.

Area of the triangle: $W = \frac{1}{2}bh$

Put in $b = x$ and $h = F_e$: $W = \frac{1}{2}xF_e$

b) Now use Hooke's law (without the negative) sub in for force in terms of k and x. Then simplify and check with your teacher

The work done to stretch the spring is stored in it as elastic potential energy.

Hooke's law: $F_e = kx$

Put it into (a): $W = \frac{1}{2}x(kx)$

Simplify: $W = \frac{1}{2}kx^2$, so $E_e = \frac{1}{2}kx^2$

FINAL ANSWER

a) $W = \frac{1}{2}xF_e$

b) $E_e = \frac{1}{2}kx^2$

EXAMPLE 1

Applying Hooke's law

A spring hangs at rest from a support. If you suspend a 0.46 kg mass from the spring, its change in length is 7.9 cm.

(a) Determine the spring constant.

(b) Calculate the change in length, in centimetres, of the same spring when a 0.75 kg mass hangs from it instead.

(c) Suppose the 0.75 kg mass is pushed upward, so that it rises past the spring's uncompressed/unstretched position, compressing the spring. Calculate the force you exert on the mass to lift it when the spring is compressed 5.3 cm above its uncompressed/unstretched length. Draw an FBD.

(d) Determine the instantaneous acceleration of the mass at the position given in (c) if you take your hand away.

GIVEN

$m_1 = 0.46$ kg

$m_2 = 0.75$ kg

$k = ?$

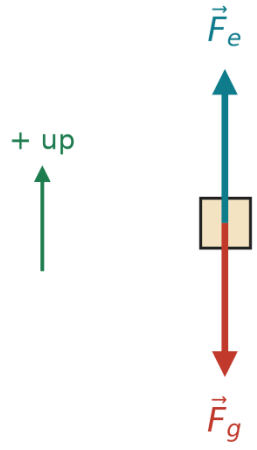
$x_1 = 7.9$ cm $= 0.079$ m

$x_c = 5.3$ cm $= 0.053$ m

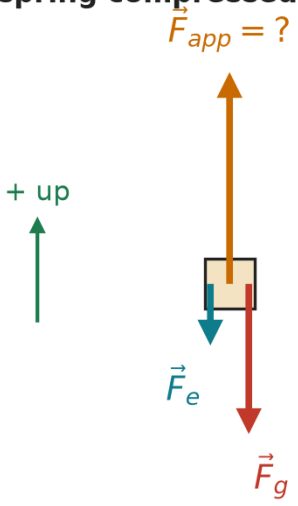
$F_{app} = ?$

FBDs of the mass (+y up)

(a), (b): mass hanging at rest



(c): spring compressed, hand holding



Hanging: the stretched spring pulls up. In (c) the spring is compressed, so it pushes down on the mass, the same way as gravity.

(a) Determine the spring constant.

The mass hangs at rest, so the forces balance. The stretched spring pulls up and gravity pulls down.

Balance: $F_{net,y} = 0$, so $0 = F_e - F_g$

Put in $F_e = kx_1$ and $F_g = m_1 g$: $kx_1 = m_1 g$

Divide both sides by x_1: $k = \dfrac{m_1 g}{x_1}$

Put in the numbers: $k = \dfrac{(0.46)(9.81)}{0.079} = 57.12 \text{ N/m}$

(b) Calculate the change in length, in centimetres, of the same spring when a 0.75 kg mass hangs from it instead.

Same balance with the new mass. Isolate the stretch this time.

Balance: $kx_2 = m_2 g$

Divide both sides by k: $x_2 = \dfrac{m_2 g}{k}$

Put in the numbers: $x_2 = \dfrac{(0.75)(9.81)}{57.12} = 0.1288 \text{ m} = 12.88 \text{ cm}$

(c) Calculate the force you exert on the mass to lift it when the spring is compressed 5.3 cm above its uncompressed/unstretched length. Draw an FBD.

The mass is held (or lifted slowly) at this position, so the forces balance. Your hand pushes up. Gravity pulls down, and the compressed spring also pushes down.

Balance: $F_{net,y} = 0, \quad \text{so} \quad 0 = F_{app} - F_g - F_e$

Add F_g and F_e: $F_{app} = m_2 g + kx_c$

Put in the numbers: $F_{app} = (0.75)(9.81) + (57.12)(0.053)$

$F_{app} = 7.3575 + 3.0274 = 10.385 \text{ N}$

(d) Determine the instantaneous acceleration of the mass at the position given in (c) if you take your hand away.

With the hand gone, gravity and the spring force both point down, so the acceleration is down. Take + down for this part (the direction of the acceleration).

Formula: $F_{net} = ma$

Put in the forces: $F_g + F_e = m_2 a$

Divide both sides by m_2: $a = \dfrac{m_2 g + kx_c}{m_2}$

Put in the numbers:

$$a = \frac{7.3575 + 3.0274}{0.75} = \frac{10.385}{0.75} = +13.85 \text{ m/s}^2$$

This is larger than g because the spring pushes down as well as gravity.

FINAL ANSWER

(a) k = 57 N/m

(b) x_2 = 13 cm

(c) $\vec{F}_{app}$ = 10. N [up]

(d) a = +14 m/s^2 (+ down)

$\vec{a}$ = 14 m/s^2 [down]

EXAMPLE 2

Calculate elastic potential energy

A 42 kg teenager balances briefly on a pogo stick, causing the spring in the stick to compress downward by 0.18 m. Determine the elastic potential energy stored in the pogo stick.

GIVEN

$m = 42$ kg $\qquad x = 0.18$ m $\qquad E_e = ?$

Step 1 – Spring constant (vertical balance)

The teenager balances, so the spring force up equals gravity down.

Balance: $F_{net,y} = 0$, so $0 = F_e - F_g$

Put in $F_e = kx$ and $F_g = mg$: $k = \dfrac{mg}{x}$

Put in the numbers: $k = \dfrac{(42)(9.81)}{0.18} = 2289.0$ N/m

Step 2 – Elastic potential energy

Formula: $E_e = \frac{1}{2}kx^2$

Put in the numbers: $E_e = \frac{1}{2}(2289.0)(0.18)^2 = 37.08\ \text{J}$

FINAL ANSWER

E_e = 37 J

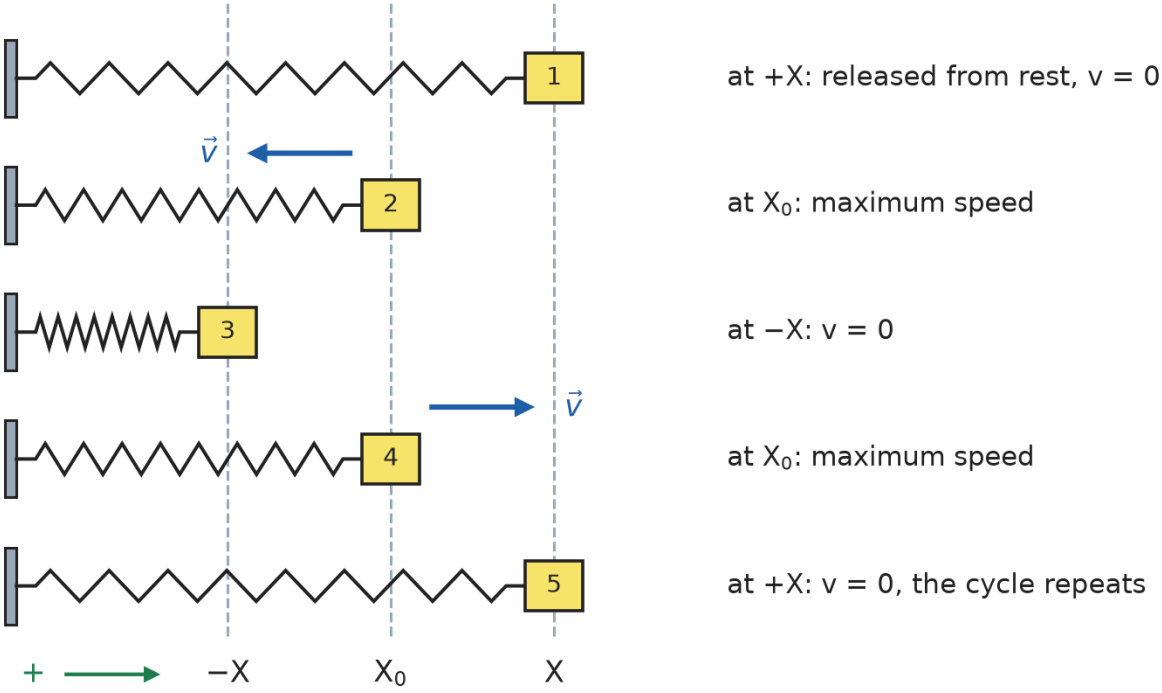
SIMPLE HARMONIC MOTION 1

Mass on a spring, one full cycle

1. Consider the spring in the diagram to the side. It sits on a frictionless surface and is pulled a distance x from its rest position. It is then released and allowed to oscillate back and forth freely. On the diagram, label the following:
    - o Increasing speed
    - o Decreasing Speed
    - o Maximum Speed
    - o Acceleration to the right
    - o Acceleration to the left
    - o Zero Acceleration
    - o Max Acceleration
    - o The type(s) of energy involved at each point
    - o When each type of energy is at its maximum
    - o Direction of the force on the mass

The spring force is a restoring force, $\vec{F}_e = -k\vec{x}$: it always points toward the rest position X_0 and is largest at the ends. The acceleration points the same way as the force. The surface is frictionless, so the total energy $E_e + E_k$ stays the same.

The five pictures on the worksheet (+ to the right)



Labels for each picture and for the motion between them:

Where	Speed	Force and acceleration	Energy
1: at +X	zero (released from rest)	maximum, to the left	all E_e (maximum)
1 → 2	increasing	to the left, getting smaller	E_e changes into E_k
2: at X_0	maximum	zero force, zero acceleration	all E_k (maximum)
2 → 3	decreasing	to the right, getting larger	E_k changes into E_e
3: at −X	zero	maximum, to the right	all E_e (maximum)
3 → 4	increasing	to the right, getting smaller	E_e changes into E_k
4: at X_0	maximum	zero force, zero acceleration	all E_k (maximum)
4 → 5	decreasing	to the left, getting larger	E_k changes into E_e
5: at +X	zero	maximum, to the left	all E_e (maximum)

FINAL ANSWER

Speed: maximum at X_0, zero at $\pm X$; increasing toward X_0, decreasing away from it

Force and acceleration: always toward X_0; zero at X_0, maximum at $\pm X$

Energy: all E_e at $\pm X$, all E_k at X_0, a mix in between

UNIT CIRCLE AND GRAPHS

Position, velocity and acceleration graphs

The motion of a mass on a spring and an object moving with uniform circular motion are very similar which results in sinusoidal graphs for the position, velocity and acceleration

a) Explain the relationship between each of these graphs in terms of both energy and forces

When	Forces	Energy
y is at a maximum or minimum (the ends)	The restoring force $F = -ky$ is largest, so the acceleration is largest. It is opposite to y: the a graph is the y graph flipped upside down.	$v = 0$, so $E_k = 0$. All the energy is potential energy.
y = 0 (equilibrium)	The net force is zero, so $a = 0$.	The speed is largest, so E_k is at its maximum and the potential energy is at its minimum.
In between	The force always points back toward $y = 0$. Moving toward $y = 0$ the mass speeds up; moving away it slows down.	Energy changes back and forth between potential and kinetic. The total stays the same.

FINAL ANSWER

a = −(k/m)y: the acceleration graph is the position graph upside down, because the force is a restoring force

The velocity graph is largest where y = 0 (all kinetic energy) and zero where y is largest (all potential energy)

ENERGY IN SIMPLE HARMONIC MOTION

Example

Example: The amplitude of vibration of a mass on a spring experiencing SHM is 0.18 m. The mass is 58 g and the spring constant is 36 N/m.

(a) Find the maximum energy of the system and the maximum speed of the mass.

(b) What amplitude of vibration would be needed to double the maximum energy?

(c) What is the maximum speed of the mass at this new energy?

GIVEN

$A = 0.18$ m $\qquad m = 58$ g $= 0.058$ kg

$k = 36$ N/m

(a) Find the maximum energy of the system and the maximum speed of the mass.

Step 1 – Total energy

At the amplitude the mass is at rest, so all of the energy is elastic potential energy.

Formula: $E_{TOT} = E_e = \frac{1}{2}kA^2$

Put in the numbers: $E_{TOT} = \frac{1}{2}(36)(0.18)^2 = 0.5832$ J

Step 2 – Maximum speed

At the equilibrium position the spring is unstretched, so all of the energy is kinetic.

Formula: $E_{TOT} = \frac{1}{2}mv_{max}^2$

Multiply by $\frac{2}{m}$ and square root: $v_{max} = \sqrt{\frac{2E_{TOT}}{m}}$

Put in the numbers: $v_{max} = \sqrt{\frac{2(0.5832)}{0.058}} = 4.484 \text{ m/s}$

(b) What amplitude of vibration would be needed to double the maximum energy?

The energy depends on A^2. Write the new energy as twice the old one and isolate the new amplitude A_2.

Double the energy: $\frac{1}{2}kA_2^2 = 2\left(\frac{1}{2}kA^2\right)$

Divide both sides by $\frac{1}{2}k$: $A_2^2 = 2A^2$

Square root both sides: $A_2 = \sqrt{2}\,A = \sqrt{2}\,(0.18) = 0.2546 \text{ m}$

(c) What is the maximum speed of the mass at this new energy?

Same as Step 2, with the doubled energy 2(0.5832) = 1.1664 J.

Formula: $v_{max} = \sqrt{\frac{2E_{TOT}}{m}}$

Put in the numbers: $v_{max} = \sqrt{\frac{2(1.1664)}{0.058}} = 6.342 \text{ m/s}$

Doubling the energy multiplies both the amplitude and the maximum speed by √2.

FINAL ANSWER

(a) E_{TOT} = 0.58 J; v_{max} = 4.5 m/s

(b) A_2 = 0.25 m

(c) v_{max} = 6.3 m/s

PERIOD OF SIMPLE HARMONIC MOTION

The restoring force in each equation

The two equations are obviously similar. You know that for simple harmonic motion to occur, there must be a restoring force.

a) What part of the spring equation relates to the restoring force on the spring? What part of the pendulum equation relates to the restoring force on the pendulum?

Spring: the spring constant k. The restoring force is F = −kx, so a stiffer spring (larger k) pulls back harder and the period T = 2π√(m/k) is shorter.
Pendulum: the gravitational field strength g. The restoring force is the part of gravity that points along the swing, back toward the lowest point. Stronger gravity pulls the bob back harder and the period T = 2π√(L/g) is shorter.

FINAL ANSWER

Spring: k (the restoring force is −kx)

Pendulum: g (the restoring force is a component of gravity)

CLASS WORK 1

Energy graph for a horizontal spring

1. The graph shows the total energy, kinetic energy and elastic potential energy for a horizontal spring of mass 0.12 kg undergoing SHM.

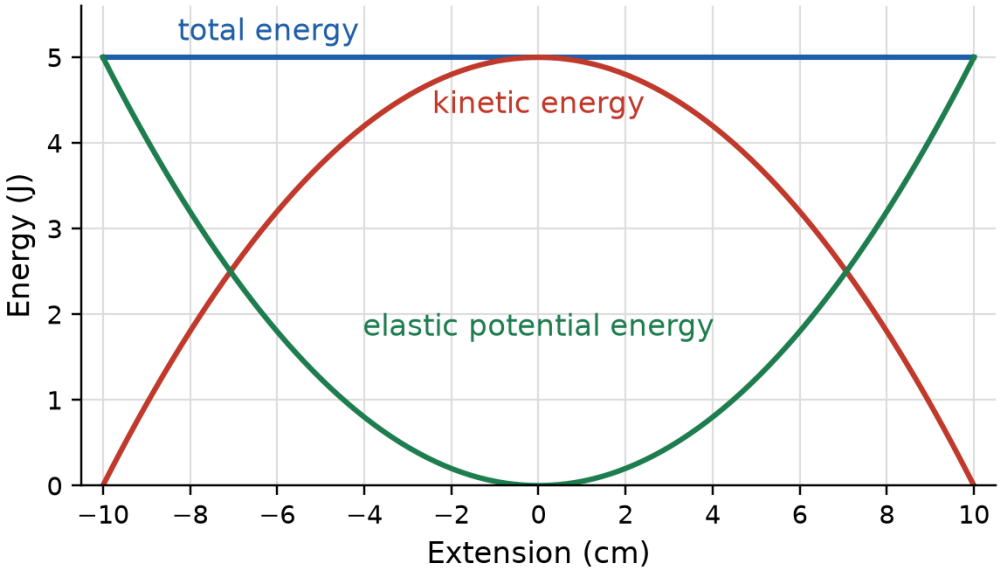
   a) Which line represents the total energy?

   b) Which line represents the kinetic energy? Which is the elastic potential energy?

   c) What is the spring constant?

   d) What is the maximum speed of the mass?

The worksheet's graph, with each line named



Read from the graph: the total energy is 5 J and the amplitude is 10 cm = 0.10 m.

a) Which line represents the total energy?

The flat line at 5 J. There is no friction, so the total energy is the same at every extension.

b) Which line represents the kinetic energy? Which is the elastic potential energy?

Elastic potential energy is the curve that is zero at 0 cm and rises to 5 J at ±10 cm, because $E_e = \frac{1}{2}kx^2$ is zero when the spring is unstretched. Kinetic energy is the curve that is 5 J at 0 cm and falls to zero at ±10 cm: the mass is fastest at equilibrium and stops at the ends.

c) What is the spring constant?

At x = 0.10 m all 5 J is elastic potential energy.

Formula: $E_e = \frac{1}{2}kx^2$

Multiply by 2 and divide by x^2: $k = \frac{2E_e}{x^2}$

Put in the numbers: $k = \frac{2(5.0)}{(0.10)^2} = 1000 \text{ N/m}$

d) What is the maximum speed of the mass?

At 0 cm all 5 J is kinetic energy.

Formula: $E_k = \frac{1}{2}mv_{max}^2$

Multiply by $\frac{2}{m}$ and square root: $v_{max} = \sqrt{\frac{2E_k}{m}}$

Put in the numbers: $v_{max} = \sqrt{\frac{2(5.0)}{0.12}} = 9.129 \text{ m/s}$

FINAL ANSWER

a) The flat line at 5 J

b) Kinetic: the arch (5 J at 0 cm). Elastic: the U-shaped curve (0 J at 0 cm)

c) $k = 1.0 \times 10^3$ N/m

d) $v_{max} = 9.1$ m/s

Worksheet answer: d) 9.13 m/s
Why the answers differ: Same value, different rounding. √(2(5.0)/0.12) = 9.129 m/s. The mass (0.12 kg) and the values read from the graph have 2 significant figures, so the key gives 9.1 m/s; the worksheet keeps 3.

CLASS WORK 2

Two students and a spring

How the question is read: in the first set-up both students pull on the free end of the spring, each with force F. In the second, one student pulls on each end, each with the same force F.

2. Two students together pull with equal force on a horizontal spring attached firmly to a wall. They then detach the spring from the wall and pull horizontally on its ends. If they each pull with the same force as before, is the amount of stretch of the spring equal to, greater than, or less than the first stretch? Explain with the use of FBDs.

Call each student's pull F. A spring at rest has zero net force, so the pulls on its two ends are equal and opposite, and that pull is what stretches it ($F_e = kx$).

FBDs of the spring in each set-up

Set-up 1: attached to the wall



Set-up 2: one student on each end



Set-up 1: both students pull on the same end, so that end is pulled with 2F, and the wall pulls back on the other end with 2F. Set-up 2: each end is pulled with only F.

$$\text{Set-up 1:}\quad 2F = kx_1 \quad\Rightarrow\quad x_1 = \frac{2F}{k}$$

$$\text{Set-up 2:}\quad F = kx_2 \quad\Rightarrow\quad x_2 = \frac{F}{k} = \frac{1}{2}x_1$$

In set-up 2, one student is only doing the job the wall did before: holding the other end still.

FINAL ANSWER

Less than the first stretch: the spring stretches half as far

CLASS WORK 3

Stretched or compressed

3. Is the amount of elastic potential energy stored in a spring greater when the spring is stretched 2.0 cm or compressed by the same amount? Explain your answer with the use of an equation.

The energy depends on x^2, so the sign of x (stretch or compression) does not matter.

$$\text{Formula:}\quad E_e = \frac{1}{2}kx^2$$

$$\text{Stretched:}\quad E_e = \frac{1}{2}k(+0.020)^2 = \frac{1}{2}k(0.00040)$$

$$\text{Compressed:}\quad E_e = \frac{1}{2}k(-0.020)^2 = \frac{1}{2}k(0.00040)$$

FINAL ANSWER

The same in both cases, because x is squared in $E_e = \frac{1}{2}kx^2$

HOMEWORK 1

Damped harmonic motion

1. Make a brief note on *damped harmonic motion* including situations where it would be preferred to SHM.

In damped harmonic motion a force such as friction or fluid resistance removes mechanical energy from the oscillating system. The amplitude gets smaller with each cycle until the object stops at its equilibrium position. The lost energy becomes thermal energy.

Damping is wanted wherever an oscillation should die out quickly: the shock absorbers of a car (so the car does not keep bouncing after a bump), door closers, bathroom scales and other measuring instruments (so the reading settles), and buildings or bridges that must not keep swaying in wind or an earthquake.

FINAL ANSWER

Damped harmonic motion: an oscillation whose amplitude shrinks because friction removes energy

Preferred when the oscillation should stop quickly, for example car shock absorbers